Fiscal Year:	FY 2011	**Task Last Updated:**	FY 05/02/2012
PI Name:	Hoffman, Larry Ph.D.		
Project Title:	Synaptic plasticity in mammalian vestibular maculae		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology (2) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	lfh@ucla.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	310-825-9823
Organization Name:	University of California Los Angeles		
PI Address 1:	Geffen School of Medicine at UCLA		
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PI Web Page:			
City:	Los Angeles	**State:**	CA
Zip Code:	90095-1624	**Congressional District:**	33
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2007 Space Flight: Bion-M1 NNH07ZTT001N
Start Date:	08/01/2009	**End Date:**	07/31/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sun, Sidney C	**Contact Phone:**	650-604-4835
Contact Email:	Sidney.C.Sun@nasa.gov		
Flight Program:	Bion/Shuttle		
Flight Assignment:	STS-131 ; Bion M1 NOTE: End date is now 7/31/2014, per A. Chu/ARC and NSSC info (Ed., 4/30/2013)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Zampighi, Guido (UCLA School of Medicine) Schietroma, Cataldo (UCLA School of Medicine) Schweizer, Felix (University of California, Los Angeles)		
Grant/Contract No.:	NNX09AL57G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The nervous systems of animals, including man, have been found to exhibit a remarkable ability to adapt to changes in the ambient gravity environment. An understanding of these adaptations is critical to managing human exploration and survivability during long-term spaceflight. Previous studies have shown that a gravity-sensing structure within the inner ear, the utricle, exhibited quantitative changes in components of its constituent hair cells during microgravity exposures up to 14 days. This structure is known as one of the vestibular maculae, inner ear structures with important load sensing capabilities. The changes that were observed involved modification in the number of synaptic ribbons, which are organelles involved in the transfer of sensory signals from the hair cells to neurons that communicate with the central nervous system. Such changes represent a form of adaptation that is more generally characterized as synaptic plasticity. At present, however, it is not known whether these changes represented a transient effect of short-term spaceflight, or whether they are more permanent and reflect a chronic adaptive modification. Therefore, the present study is designed to test whether the approximately 30-day exposure to microgravity proposed for the BION-M1 satellite mission results in chronic modification in synapse number within hair cells of the utricle. This will be achieved through transmission electron microscopy and quantitative analyses to estimate the density of synaptic ribbons in different hair cell types and in different regions of utricles from flight and control animals. In addition, we will utilize new methods of conical tomography to visualize these ribbon synapses in three dimensions. This enables us to test whether synaptic plasticity is manifested in changes in subpopulations of ultrastructural elements of the synaptic ribbon. These results will be critical to a more comprehensive understanding of the long-term exposure to spaceflight in mammalian gravity-sensing systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	[Editor's note 5/02/2012: No Task Book report received. Progress section and Bibliography compiled from PI's Annual Technical Report.] We are conducting ultrastructural analyses of the inner ear utricle, testing the hypothesis that exposure to microgravity incumbent with spaceflight results in modifications in the density of synapses within hair cells of the utricle, and in modifications within the synapse, per se. These latter modifications will be in the compartments of neurotransmitter vesicles associated with each synapse. Specimens were obtained from mice exposed to microgravity aboard STS-131 (April 2010), and from control specimens. Temporal bone specimens were obtained shortly after landing at Kennedy Space Center. These were shipped to the PI's laboratory at UCLA, where they were microdissected to remove the inner ear vestibular epithelia. They were then processed for transmission electron microscopy. Ultrastructural analyses will be completed that will produce estimates of synaptic ribbon density (ribbons per hair cell). Conical electron tomography will also be conducted in order to obtain three-dimensional reconstructions for quantitative analyses of the pools of neurotransmitter vesicles. Additional specimens obtained from STS-131 have enabled us to complete a immunohistochemical analysis of synapse density, which will complement our ultrastructural findings. During this reporting period we participated in the BSP programs associated with STS-133 and STS-135. The BSP adjoined to STS-133 involved very few specimens due to the parsing of flight animals by the principal investigator team. These specimens have been prepared for ultrastructural processing, which is in progress. The BSP associated with STS-135 that were distributed to our team were compromised and we are unable to utilize them.
Bibliography Type:	Description: (Last Updated: 10/05/2022)
Abstracts for Journals and Proceedings	Sultemeier DR, Hoffman LF, Quinones PM, Schweizer F. "Sensory learning during spaceflight: alterations in synapse density following exposure to microgravity." Midwinter meeting of the Association for Research in Otolaryngology, Baltimore, MD, February 19-23, 2011. Association for Research in Otolaryngology Abstracts: 255. http://www.aro.org/archives/2011/2011_255_1305771726.html ; http://www.aro.org/mwm/documents/2011_Abstract_Book.pdf , Feb-2011
Abstracts for Journals and Proceedings	Hoffman LF, Sultemeier DR, Quinones PM, Zampighi GZ, Schweizer F. "STS-131: Spaceflight-induced Plasticity of Utricular Hair Cell Synapses." Eighth Symposium on the Role of the Vestibular Organs in Space Exploration, Houston, TX, April 8-10, 2011. Eighth Symposium on the Role of the Vestibular Organs in Space Exploration, Houston, TX, April 8-10, 2011. http://www.dsls.usra.edu/meetings/IAA/vestibular/pdf/3045.pdf , Apr-2011
Abstracts for Journals and Proceedings	Hoffman LF, Sultemeier DR, Quinones PM, Zampighi GZ, Schweizer F. "STS-131: Sensory Learning During Spaceflight. Alterations in Utricular Hair Cell Synapse Density Following Exposure to Microgravity." 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. , Apr-2011